

High-speed droplet-allele-specific polymerase chain reaction for genotyping of single nucleotide polymorphisms.

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Single nucleotide alternations such as single nucleotide polymorphisms (SNPs) or single nucleotide mutations are useful genetic markers for molecular diagnosis, prognosis, drug response, and predisposition to diseases. Rapid identification of SNPs or mutations is clinically important, especially for determining drug responses and selection of molecular-targeted therapy. Here, we describe a rapid genotyping assay based on the allele-specific polymerase chain reaction (AS-PCR) by using our droplet-PCR machine (droplet-AS-PCR).

New complementary python codes to locate Single Nucleotide Polymorphisms (SNPs) and Overlapping G-Quadruplex Sequences (G4s).

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G-quadruplexes (G4s) are non-canonical DNA and RNA secondary structures that control gene regulation. A single nucleotide polymorphism (SNP) is a small genetic variation occurring within a DNA sequence and accounting for the variabilities between individuals. While the majority of SNPs, especially those frequent in the population, are considered as benign genetic variations, few others can lead to diseases. SNPs occurring in G4 sequences were reported to modulate gene regulation. In order to find overlaps between predicted G4 sequences and SNPs located in the genomic regions, we developed two complementary computational python codes (SNP-locator and G4-overlap). The codes map a mutation to the overlapping/closest G4 sequences, based on the genetic variant name and the FASTA format of the corresponding gene. We validated these two codes on a set of 31 SNP variants occurring in cytochromes P450 genes and podocytes-marker genes. Out of 31 SNPs, 28 were accurately located using the mentioned codes. •SNP-locator code locates any SNP in promoters, upstream regulatory regions, exons and introns. •The SNP-locator code requires the FASTA genomic sequence of the studied gene and the genetic variant nomenclature at the cDNA level. •G4-overlap code maps the SNP to the overlapping or the closest G4 sequence.

Heme modulates Trypanosoma cruzi bioenergetics inducing mitochondrial ROS production.

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Trypanosoma cruzi is the causative agent of Chagas disease and has a single mitochondrion, an organelle responsible for ATP production and the main site for the formation of reactive oxygen species (ROS). T. cruzi is an obligate intracellular parasite with a complex life cycle that alternates between vertebrate and invertebrate hosts, therefore the development of survival strategies and morphogenetic adaptations to deal with the various environments is mandatory. Over the years our group has been studying the vector-parasite interactions using heme as a physiological oxidant molecule that triggered epimastigote proliferation however, the source of ROS induced by heme remained unknown. In the present study we demonstrate the involvement of heme in the parasite mitochondrial metabolism, decreasing oxygen consumption leading to increased mitochondrial ROS and membrane potential. First,

we incubated epimastigotes with carbonyl cyanide p-(trifluoromethoxy) phenylhydrazone (FCCP), an uncoupler of oxidative phosphorylation, which led to decreased ROS formation and parasite proliferation, even in the presence of heme, correlating mitochondrial ROS and *T. cruzi* survival. This hypothesis was confirmed after the mitochondria-targeted antioxidant ((2-(2,2,6,6-Tetramethylpiperidin-1-oxyl-4-ylamino)-2-oxoethyl) triphenylphosphonium chloride (MitoTEMPO) decreased both heme-induced ROS and epimastigote proliferation. Furthermore, heme increased the percentage of tetramethylrhodamine methyl ester (TMRM) positive parasites tremendously-indicating the hyperpolarization and increase of potential of the mitochondrial membrane ($\Delta\Psi_m$). Assessing the mitochondrial functional metabolism, we observed that in comparison to untreated parasites, heme-treated epimastigotes decreased their oxygen consumption, and increased the complex II-III activity. These changes allowed the electron flow into the electron transport system, even though the complex IV (cytochrome c oxidase) activity decreased significantly, showing that heme-induced mitochondrial ROS appears to be a consequence of the enhanced mitochondrial physiological modulation. Finally, the parasites that were submitted to high concentrations of heme presented no alterations in the ultrastructure. Consequently, our results suggest that heme released by the insect vector after the blood meal, modify epimastigote mitochondrial physiology to increase ROS as a metabolic mechanism to maintain epimastigote survival and proliferation.

Hormonal orchestra: mastering mitochondria's role in health and disease.

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Mitochondria is a subcellular organelle involved in the pathogenesis of cellular stress, immune responses, differentiation, metabolic disorders, aging, and death by regulating process of fission, fusion, mitophagy, and transport. However, an increased interest in mitochondria as powerhouse for ATP production, the mechanisms of mitochondria-mediated cellular dysfunction in response to hormonal interaction remains unknown. Mitochondrial matrix contains chaperones and proteases that regulate intrinsic apoptosis pathway through pro-apoptotic Bcl-2 family's proteins Bax/Bak, and Cyt C release, and induces caspase-dependent and independent cells death. Energy and growth regulators such as thyroid hormones have profound effect on mitochondrial inner membrane protein and lipid compositions, ATP production by regulating oxidative phosphorylation system. Mitochondria contain cholesterol side-chain cleavage enzyme, P450_{scc}, ferredoxin, and ferredoxin reductase providing an essential site for steroid hormones biosynthesis. In line with this, neurohormones such as oxytocin, vasopressin, and melatonin are correlated with mitochondrial integrity, displaying therapeutic implications for inflammatory and immune responses. Melatonin's also displayed protective role against oxidative stress and mitochondrial synthesis of ROS, suggesting a defense mechanism against aging-related diseases. An imbalance in mitochondrial bioenergetics can cause neurodegenerative disorders, cardiovascular diseases, and cancers. Hormone-induced PGC-1 α stimulates mitochondrial biogenesis via activation of NRF1 and NRF2, which in turn triggers mtTFA in brown adipose and cardiac myocytes. Mitochondria can be transferred through cells merging, exosome-mediated transfer, and tunneling through nanotubes. By delineating the underlying molecular mechanism of hormonal mitochondrial interaction, this study reviews the dynamics mechanisms of mitochondria and its effects on cellular level, health, diseases, and therapeutic strategies targeting mitochondrial diseases.

Mitochondrial Structure and Bioenergetics in Normal and Disease Conditions.

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Mitochondria are ubiquitous intracellular organelles found in almost all eukaryotes and involved in various aspects of cellular life, with a primary role in energy production. The interest in this organelle has grown stronger with the discovery of their link to various pathologies, including cancer, aging and neurodegenerative diseases. Indeed, dysfunctional mitochondria cannot provide the required energy to tissues with a high-energy demand, such as heart, brain and muscles, leading to a large spectrum of clinical phenotypes. Mitochondrial defects are at the origin of a group of clinically heterogeneous pathologies, called mitochondrial diseases, with an incidence of 1 in 5000 live births. Primary mitochondrial diseases are associated with genetic mutations both in nuclear and mitochondrial DNA (mtDNA), affecting genes involved in every aspect of the organelle function. As a consequence, it is difficult to find a common cause for mitochondrial diseases and, subsequently, to offer a precise clinical definition of the pathology. Moreover, the complexity of this condition makes it challenging to identify possible therapies or drug targets.

Mitochondria-targeted drug delivery in cancers.

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Mitochondria are considered one of the most important subcellular organelles for targeting and delivering drugs because mitochondria are the main location for various cellular functions and energy (i.e., ATP) production, and mitochondrial dysfunctions and malfunctions cause diverse diseases such as neurodegenerative disorders, cardiovascular disorders, metabolic disorders, and cancers. In particular, unique mitochondrial characteristics (e.g., negatively polarized membrane potential, alkaline pH, high reactive oxygen species level, high glutathione level, high temperature, and paradoxical mitochondrial dynamics) in pathological cancers have been used as targets, signals, triggers, or driving forces for specific sensing/diagnosing/imaging of characteristic changes in mitochondria, targeted drug delivery on mitochondria, targeted drug delivery/accumulation into mitochondria, or stimuli-triggered drug release in mitochondria. In this review, we describe the distinctive structures, functions, and physiological properties of cancer mitochondria and discuss recent technologies of mitochondria-specific "key characteristic" sensing systems, mitochondria-targeted "drug delivery" systems, and mitochondrial stimuli-specific "drug release" systems as well as their strengths and weaknesses.

Mitochondrial ATP synthases cluster as discrete domains that reorganize with the cellular demand for oxidative phosphorylation.

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Mitochondria are double membrane-bounded organelles that form a dynamic tubular network. Mitochondria energetic functions depend on a complex internal architecture. Cristae, inner membrane invaginations that fold into the matrix space, are proposed to be the site of oxidative phosphorylation, reactions by which ATP synthase produces ATP. ATP synthase is also thought to have a role in crista morphogenesis. To date, the exploration of the processes regulating mitochondrial internal compartmentalization have been mostly limited to electron microscopy. Here, we describe ATP synthase localization in living yeast cells and show that it clusters as discrete inner membrane domains. These domains are dynamic within the mitochondrial network. They are impaired in mutants defective in crista

morphology and partially overlap with the crista-associated MICOS-MINOS-MITOS complex. Finally, ATP synthase occupancy increases with the cellular demand for OXPHOS. Overall our data suggest that domains in which ATP synthases are clustered correspond to mitochondrial cristae. Being able to follow mitochondrial sub-compartments in living yeast cells opens new avenues to explore the mechanisms involved in inner membrane remodeling, an architectural feature crucial for mitochondrial activities.

Mitochondrial-derived vesicles retain membrane potential and contain a functional ATP synthase.

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Vesicular transport is a means of communication. While cells can communicate with each other via secretion of extracellular vesicles, less is known regarding organelle-to organelle communication, particularly in the case of mitochondria. Mitochondria are responsible for the production of energy and for essential metabolic pathways in the cell, as well as fundamental processes such as apoptosis and aging. Here, we show that functional mitochondria isolated from *Saccharomyces cerevisiae* release vesicles, independent of the fission machinery. We isolate these mitochondrial-derived vesicles (MDVs) and find that they are relatively uniform in size, of about 100 nm, and carry selective protein cargo enriched for ATP synthase subunits. Remarkably, we further find that these MDVs harbor a functional ATP synthase complex. We demonstrate that these vesicles have a membrane potential, produce ATP, and seem to fuse with naive mitochondria. Our findings reveal a possible delivery mechanism of ATP-producing vesicles, which can potentially regenerate ATP-deficient mitochondria and may participate in organelle-to-organelle communication.

How does density of the inner mitochondrial membrane influence mitochondrial performance?

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Our current understanding of variation in mitochondrial performance is incomplete. The production of ATP via oxidative phosphorylation is dependent, in part, on the structure of the inner mitochondrial membrane. Morphology of the inner membrane is crucial for the formation of the proton gradient across the inner membrane and, therefore, ATP synthesis. The inner mitochondrial membrane is dynamic, changing shape and surface area. These changes alter density (amount per volume) of the inner mitochondrial membrane within the confined space of the mitochondrion. Because the number of electron transport system proteins within the inner mitochondrial membrane changes with inner mitochondrial membrane area, a change in the amount of inner membrane alters the capacity for ATP production within the organelle. This review outlines the evidence that the association between ATP synthases, inner mitochondrial membrane density, and mitochondrial density (number of mitochondria per cell) impacts ATP production by mitochondria. Furthermore, we consider possible constraints on the capacity of mitochondria to produce ATP by increasing inner mitochondrial membrane density.

Measuring Mitochondrial Function: From Organelle to Organism.

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Mitochondrial energy production is crucial for normal daily activities and maintenance of life. Herein, the logic and execution of two main classes of measurements are outlined to delineate mitochondrial function: ATP production and oxygen consumption. Aerobic ATP production is quantified by phosphorus magnetic resonance spectroscopy ($^{31}\text{PMRS}$) in vivo in both human subjects and animal models using the same protocols and maintaining the same primary assumptions. Mitochondrial oxygen consumption is quantified by oxygen polarography and applied in isolated mitochondria, cultured cells, and permeabilized fibers derived from human or animal tissue biopsies. Traditionally, mitochondrial functional measures focus on maximal oxidative capacity—a flux rate that is rarely, if ever, observed outside of experimental conditions. Perhaps more physiologically relevant, both measurement classes herein focus on one principal design paradigm; submaximal mitochondrial fluxes generated by graded levels of ADP to map the function for ADP sensitivity. We propose this function defines the bioenergetic role that mitochondria fill within the myoplasm to sense and match ATP demands. Any deficit in this vital role for ATP homeostasis leads to symptoms often seen in cardiovascular and cardiopulmonary diseases, diabetes, and metabolic syndrome.
